RESEARCH DESIGN / BAYESIAN SOURCE SEPARATION

Score-based inverse modelling for NILM

Non-intrusive load monitoring (NILM) with one total meter, five target appliances and one unmonitored background

01 TRAINING
Learn source priors

ALIGNED TRAINING WINDOWS

Use individual power windows from the fridge, microwave, kettle, washer and dishwasher.

Use the training meter to obtain the effective background.

$$\boldsymbol{r}_{\text{train}} = \boldsymbol{y}_{\text{train}} - \sum_{i=1}^{5} \boldsymbol{p}_{i,\text{label}}$$

ADD NOISE TO ONE SOURCE AT A TIME

$$\tilde{\boldsymbol{v}} = \boldsymbol{v} + \sigma\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(0, I)$$

1D CNN SCORE NETWORK

Inputs: noisy window, source ID j and noise level σ. Fit the target score.

$$\mathcal{L} = \left\| S_\theta(\tilde{\boldsymbol{v}}, j, \sigma) - \frac{\boldsymbol{v} - \tilde{\boldsymbol{v}}}{\sigma^2} \right\|_2^2.$$

02 TEST-TIME INVERSE INFERENCE
Sample a decomposition

INPUT ONE OBSERVED METER WINDOW y_{obs}

Initialise six candidate curves $\boldsymbol{x}^{(0)} = (\boldsymbol{p}_1, \ldots, \boldsymbol{p}_5, \boldsymbol{r})^{(0)}$.

1. PREDICT THE TOTAL WITH KNOWN POWER BALANCE

$$\hat{\boldsymbol{y}}^{(k)} = F(\boldsymbol{x}^{(k)}) = \textstyle\sum_{i=1}^{5} \boldsymbol{p}_i^{(k)} + \boldsymbol{r}^{(k)}$$

2. COMBINE METER AGREEMENT AND SOURCE PRIORS

$$\log p_\ell(\boldsymbol{x} \mid \boldsymbol{y}_{\text{obs}}) = -\frac{\|F(\boldsymbol{x}) - \boldsymbol{y}_{\text{obs}}\|_2^2}{2\gamma_\ell^2} + \textstyle\sum_{j=1}^{6} \log q_{j,\sigma_\ell}(\boldsymbol{v}_j) + C_\ell$$

Independent priors before observing the meter. C_ℓ is constant in $\boldsymbol{x}$.

3. UPDATE EVERY CANDIDATE SOURCE j

$$\boldsymbol{v}_j^{(k+1)} = \boldsymbol{v}_j^{(k)} + \eta_\ell \Bigg[\underbrace{S_\theta(\boldsymbol{v}_j^{(k)}, j, \sigma_\ell)}_{\text{learned source direction}} + \underbrace{\frac{\boldsymbol{y}_{\text{obs}} - \hat{\boldsymbol{y}}^{(k)}}{\gamma_\ell^2}}_{\text{meter correction}} \Bigg] + \sqrt{2\eta_\ell}\, \boldsymbol{\xi}_j^{(k)}, \qquad \boldsymbol{\xi}_j^{(k)} \sim \mathcal{N}(0, I) \quad \text{(random step)}.$$

REPEAT

Anneal from high to low noise. Keep θ fixed and update the six curves.

03 OUTPUTS
Read out power and states

EACH COMPLETED SAMPLING RUN

One plausible decomposition with five target power curves and one effective background curve.

$$\hat{\boldsymbol{x}}^{(s)} = (\hat{\boldsymbol{p}}_1^{(s)}, \ldots, \hat{\boldsymbol{p}}_5^{(s)}, \hat{\boldsymbol{r}}^{(s)})$$

CONVERT POWER TO ON/OFF LABELS

$$a_i^{(s)}(t) = \mathbf{1}\big[\hat{p}_i^{(s)}(t) > \tau_i\big]$$

Apply a device threshold τ_i to each sampled curve. Multiple devices can be on at the same time.

ESTIMATE ON PROBABILITY

$$\widehat{\Pr}(a_i(t) = 1 \mid \boldsymbol{y}_{\text{obs}}) = \frac{1}{S} \textstyle\sum_{s=1}^{S} a_i^{(s)}(t)$$

Microwave on in 75 of 100 samples gives an estimated on probability of 75%.

WORKED EXAMPLE TWO ALLOCATIONS CAN EXPLAIN THE SAME METER

Observed window $[400, 2000, 400]$ W. The other three targets are off in both examples.

Candidate	Fridge	Microwave	Background	Predicted meter
A	$[100, 100, 100]$	$[0, 1500, 0]$	$[300, 400, 300]$	$[400, 2000, 400]$
B	$[100, 100, 100]$	$[0, 0, 0]$	$[300, 1900, 300]$	$[400, 2000, 400]$

WHAT DRIVES THE NEXT UPDATE?

If a current guess sums to 1900 W at the middle point, the meter correction pushes all six sources upward there. The source scores can oppose a fridge spike and support a microwave-shaped event.

A and B have equal meter fit. The priors influence their relative support. Held-out appliance labels test correct attribution.

Proposed NILM adaptation · Illustrative values, no experimental results

Sources Jayaram & Thickstun, ICML 2020 (BASIS) · Song & Ermon, NeurIPS 2019 (score matching)